One can conceive of a variety of different kinds of spacetime singularities, each with its own peculiar form of tidal stretch and squeeze, and we shall meet several different kinds in this chapter.

The singularity predicted by the Oppenheimer–Snyder calculations is a very simple one. Its tidal gravity has essentially the same form as the Earth's or Moon's or Sun's; that is, the same form as the tidal gravity that creates the tides on the Earth's oceans (Box 2.5): The singularity stretches all objects radially (in the direction toward and away from itself), and squeezes all objects transversely.

Imagine an astronaut falling feet first into the kind of black hole described by Oppenheimer and Snyder's equations. The larger the hole, the longer he can survive, so for maximum longevity, let the hole be among the largest that inhabit the cores of quasars (Chapter 9): 10 billion solar masses. Then the falling astronaut crosses the horizon and enters the hole about 20 hours before his final death, but as he enters, he is still too far from the singularity to feel its tidal gravity. As he continues to fall faster and faster, coming closer and closer to the singularity, the tidal gravity grows stronger and stronger until, just 1 second before the singularity, he begins to feel it stretching his feet and head apart and squeezing him from the sides (bottom picture in Figure 13.1). At first, the stretch and squeeze are only mildly annoying, but they continue to grow until, a few hundredths of a second before the singularity (middle picture), they get so strong that his bones and flesh can no longer resist. His body comes apart and he dies. In the last hundredth second, the stretch and squeeze continue mounting, and as he reaches the singularity, they become infinitely strong, first at his feet, then at his trunk, then at his head; his body gets infinitely distended; and then, according to general relativity, he merges with and becomes part of the singularity.

It is utterly impossible for the astronaut to move on through the singularity and come out the other side because, according to general relativity, there is no "other side." Space, time, and spacetime cease to exist at the singularity. The singularity is a sharp edge, much like the edge of a sheet of paper. There is no paper beyond its edge; there is no spacetime beyond the singularity. But there the similarity ends. An ant on the paper can go right up to the edge and then back away, but nothing can back away from the singularity; all astronauts, particles, waves, whatever, that hit it are instantaneously destroyed, according to Einstein's general relativistic laws.